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Iniziato	martedì, 15 giugno 2021, 21:35
Stato	Completato
Terminato	mercoledì, 16 giugno 2021, 12:07
Tempo impiegato	14 ore 32 min.
Valutazione	Non ancora valutato

Domanda **1**

Completo

Punteggio
max.: 1,00

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$L = \{w \in \{0,1\}^* \mid \text{\texttt{\$w\$ does not contain \$01\$ as a substring}}\}$$

Study the complexity of TM you have defined.

```
(q_init, ▷) → (q_0, ▷, R)
(q_0, □) → (q_halt, □, S)
(q_0, 0) → (q_1, 0, R)
(q_0, 1) → (q_2, 1, R)
(q_1, 0) → (q_1, 0, R)
(q_1, □) → (q_halt, □, S)
(q_2, 0) → (q_1, 0, R)
(q_2, 1) → (q_2, 1, R)
(q_2, □) → (q_halt, □, S)
```

The TM above takes linear time in the input length, as it scans the input string from the beginning until—at most—its very end.

Domanda **2**

Completo

Punteggio
max.: 1,00

You are required to prove that the following function f is in **FP**. To do that, you can give a TMs or define some pseudocode. The function is one that, given a list $L = L_1, \dots, L_n$, returns its inverse, namely $L_n, L_{n-1}, \dots, L_1$.

2-tape TM:

First tape starts as $\triangleright$ followed by the input string, i.e. $\sqcup L \sqcup = \sqcup (\sqcup L_1 \sqcup \# \dots \# \sqcup L_n \sqcup) \sqcup$, followed by infinite $\square$ s

Second tape starts as $\triangleright$ followed by infinite $\square$ s

```
(q_init,  $\triangleright$ ,  $\triangleright$ )  $\rightarrow$  (q_0,  $\triangleright$ ,  $\triangleright$ , R, R)
(q_0, 0,  $\square$ )  $\rightarrow$  (q_0, 0,  $\square$ , R, S)    // go to the end ...
(q_0, 1,  $\square$ )  $\rightarrow$  (q_0, 1,  $\square$ , R, S)    // ... of the string
(q_0,  $\square$ ,  $\square$ )  $\rightarrow$  (q_1,  $\square$ ,  $\square$ , L, S) // you reached the end
(q_1, 0,  $\square$ )  $\rightarrow$  (q_1, 0, 0, L, R)      // copy the bits ...
(q_1, 1,  $\square$ )  $\rightarrow$  (q_1, 1, 1, L, R)      // ... in the second tape
(q_1,  $\triangleright$ ,  $\square$ )  $\rightarrow$  (q_halt,  $\triangleright$ ,  $\square$ , S, S) // no more bits to copy
```

The above TM scans the whole string from beginning to end and then goes back to where it started, while copying on the second tape the symbols it reads on the first tape during the backward pass. Therefore, the TM requires $O(2l)$ steps, where l is the input encoding length, hence it works in linear time.

Since a single tape TM can simulate a 2-tape one with polynomial overhead, we can say that f is in FP.

Domanda **3**

Completo

Punteggio
max.: 1,00

We studied the problem **CLIQUE**. You are required to classify the subset **THREECLIQUE** of **CLIQUE** consisting of all the pairs $(G, 3)$. To which class does **THREECLIQUE** belong?

INPUT: $G=(V,E)$ undirected graph
 OUTPUT: 1 if G has a 3-clique, 0 otherwise

```

-----
foreach {u,v,w} in V do
  if {u,v} in E and {v,w} in E and {w,u} in E then
    return 1
return 0
  
```

1. INPUT ENCODING

The usual encoding $\langle (|V|, A) \rangle$, where A is the adjacency matrix, such that

$$I = \langle (|V|, A) \rangle = O(|V|^2)$$

which is polynomial in $|V|$.

2. NUMBER OF INSTRUCTIONS

The whole algorithm basically consists of a single foreach loop, which is executed at most as many times as there are subsets of V of cardinality 3, i.e.

$$\begin{aligned} \text{binom}(|V|, 3) &= |V|! / (3!(|V|-3)!) = (|V|(|V|-1)(|V|-2)) / 6 \\ &= O(|V|^3) = O(|V|^4) = O(I^2) \end{aligned}$$

which is polynomial in I .

3. EXECUTION TIME OF EACH INSTRUCTION

The only nontrivial instruction is checking whether a couple of nodes is connected by an arc, which can be done in time

$$O(|E|) = O(|V|^2) = O(I)$$

which is polynomial in I .

4. SIZE OF INTERMEDIATE RESULTS

The only intermediate datum is the set $\{u,v,w\}$, which has constant size 3.

◀ [Question Examples](#)

Vai a...